

Remote Online Shopping System with 360-Degree Streaming and Spherical Display

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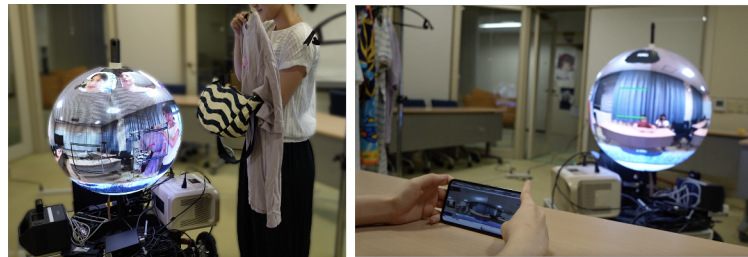
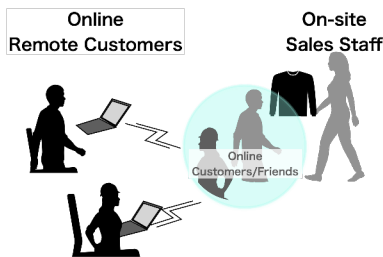


Figure 1: Remote Online Shopping System with 360-Degree Streaming and Spherical Display Overview: (Left) Concept of the system. (Middle) on-site sales staff is presenting products while observing the instructions and facial expressions of the online remote customers. (Right) The remote customer is operating the spherical interface from their own device. In actual situation, this operation is performed from a remote location where the spherical interface is not physically visible.

ABSTRACT

In traditional on-site shopping, communication with sales staff and interactions with friends or family members play a major role in selecting products and sometimes lead to unexpected discoveries. On the other hand, in online shopping, the lack of communication often leads consumers to make purchases with a predetermined mindset. This is due to the tendency to select the same products or rely on recommendation algorithms, which reduce opportunity for spontaneous product discovery. To address this, we propose a remote online shopping interface that offers an experience similar to shopping in physical stores, even for online users, by utilizing 360-degree streaming and a spherical display located in the store. We implemented a prototype and present a demonstration. The demonstration shows that the 360-degree footage displayed on the spherical screen made it easier for sales staff to understand the customer's focus and intent.

Index Terms: Remote collaboration, spherical display, online shopping.

1 INTRODUCTION

On-site shopping often involves face-to-face communication between the seller and customers, and sometimes conversations with friends or family who accompany the shopper can lead to purchase decisions. One of the joys of shopping with others is the unexpected moment when a friend, who may have different tastes, suggests a

product that turns out to be surprisingly appealing. In contrast, online shopping is typically a solitary activity. While automated suggestions for related products are available, shoppers miss out on the explanations that would normally arise from communication with store staff, such as why certain items are recommended. This can lead to a narrowing of product choices, which is detrimental to both customers and sellers.

Even in online shopping, it is desirable to engage in conversation with store staff or companions, and to pay attention to surrounding products, in a manner similar to that observed in offline shopping. One potential solution to enhance online shopping is the use of remote robots [2, 7, 8]. However, this approach faces limitations such as a restricted field of view and limited gesture range. Beyond shopping, the issue of a narrow field of view in remote communication has been addressed by using 360-degree cameras [1, 4, 6, 9]. In these studies, remote users observe the 360-degree video feed from the local site, allowing them to provide instructions and communicate, but with limited viewing angle or lack of visual cues due to display methods. In contrast, there are studies on remote collaborative work that utilize spherical displays to accurately convey instructions, including 360-degree surrounding information from both the local and remote sides [3, 5, 10]. Similarly, this research aims to improve remote communication by equipping a remotely operable wheeled robot with a spherical display, enabling accurate transmission of directional information.

To provide an online shopping experience where users can communicate with friends and family while shopping, we propose a remote shopping system with 360-degree streaming and a spherical display. This system offers two-way communication through utilizing a 360-degree camera and a maneuverable spherical display, as shown in Figure 1. The system is installed within the store which allows customers to engage in remote online shopping by observing the 360-degree video feed. The spherical display mirrors the 360-degree view of the store, with the online shopper's face and focal point superimposed on it. By augmenting the spatial environment with 360-degree video and spherical displays, directional information can be communicated more effectively. This enables customers to indicate their points of interest to the store staff, who

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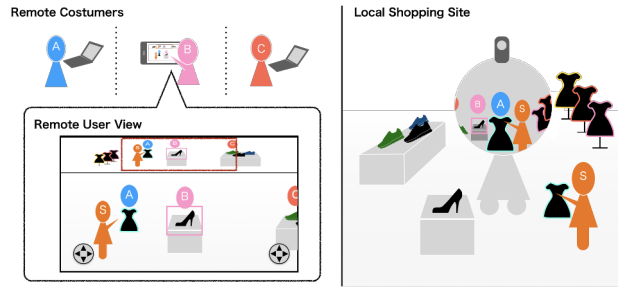


Figure 2: The concept of the remote online shopping with 360-degree streaming and spherical display: (Left) Remote customers Side: Multiple remote customers connect to on-site 360-degree view and interact with other users and on-site sales staff via the system. (Right) Local Shopping site: A spherical interface moves around in the shopping site controlled by a remote customer. On-site sales staff promote products while observing remote customers' faces and points of interest via the system.

can observe the customer's expressions and focus points displayed on the sphere, facilitating appropriate and responsive service. In this paper, we prototype the system and validate it through a demonstration.

2 PROTOTYPE OF REMOTE ONLINE SHOPPING SYSTEM WITH 360-DEGREE STREAMING AND SPHERICAL DISPLAY

We propose this shopping system as a new remote shopping style which provides remote users with the experience of shopping locally together. Figure 2 shows the overview of our system.

2.1 Concept of the system

In this system, a spherical interface is installed at the physical store, including a maneuverable spherical display equipped with a 360-degree camera. Remote users are able to engage in e-commerce activities by observing the 360-degree video captured via their own devices through a server connection. This system provides the following experiences to remote consumers and local sellers.

Remote Consumer: The remote consumers can access the Spherical Shopping app from their PC or tablet. The app displays the 360-degree video feed from the store at the top of the screen, while the user's 120-degree field of view is shown across the entire display. The user can adjust the angle of their field of view using the control joystick at the bottom right of the app (see Figure 2 Remote user view). Additionally, they can remotely control the physical spherical display at the store via the control joystick at the bottom left, allowing them to navigate the store as if they were walking through it in person. When the user taps on the screen, the tapped area is highlighted with a square, enabling them to communicate their point of interest to other remote users and local seller.

Local Seller: On the spherical display at the on-site store, the 360-degree camera view is shown like a mirror. The faces of remote users are displayed above their respective points of focus. We found this location to be very suitable since by the nature of the 360-degree video the upper area consists of the ceiling of the store, not showing any products. When a remote user taps on an area of interest, a square marker appears on the sphere, indicating the direction of the product they are focusing on.

With these features, remote users can shop while conversing with other remote users and store staff. They can point to products and give instructions to the staff or align their actions with their facial

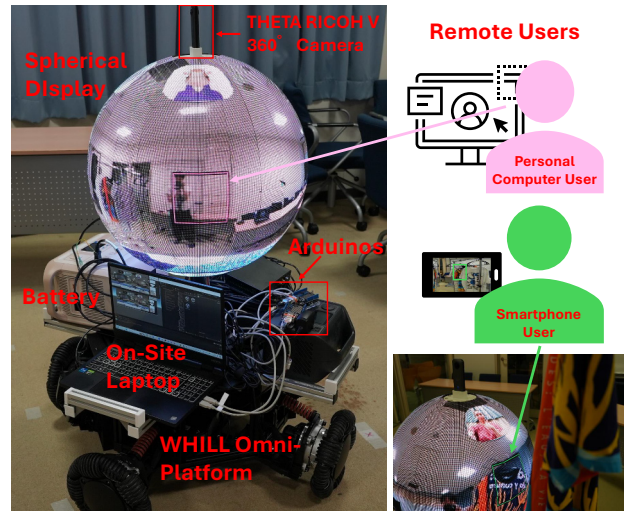


Figure 3: Hardware overview consisting of the proposed system. The System can be accessed and controlled with both smartphones and personal computers.

display, simulating an in-person shopping experience. Additionally, the store staff can observe the remote users' gestures and facial expressions to provide accurate product recommendations and purchasing advice.

2.2 Implementation

In general, there are two main sites: The store site with the store's tenant that can provide support and the customer's site with the remote users of the system that can participate in on-site shopping from a location of their choice.

The system's hardware consists of the following components which can be seen in Figure 3: A RICOH THETA V¹ 360-degree camera is on the top of a spherical display to capture the store site's surroundings for the remote customer. A Spherical display is used on the store site to display the 360-degree image with the remote customer's point of interest displayed for the store tenant. Portable batteries are used to power both the spherical display and the laptop. WHILL Omni-Platform² is referred to as OmniWheel to transport the store site's components such as the spherical display, battery, and laptop. The platform can be moved and rotated via two individual, crossed axes with wheels at their end. Both axes are split in the center, allowing the wheels to move in the same direction (moving the platform) or opposite directions (rotating the platform). Two Arduinos are to control the OmniWheel's two steering axes. A Laptop on the spherical display is used to capture and distribute the camera image, to receive and forward the OmniWheel control to the Arduino, and to display the user interface for the spherical display. Devices are on the customers' sites to display the store's 360-degree view and to send the user's point of interest. Currently, our system supports multiple devices from smartphones for both Android and iOS to computers. For our user test, we provided 2 Android smartphones.

3 DEMONSTRATION

We conducted a preliminary demonstration in our laboratory as a proof of concept. We prepared one on-site sales staff and two remote customers: one acting as the primary shopper looking for

¹<https://www.ricoh360.com/ja/theta/>

²<https://whill.inc/jp/>

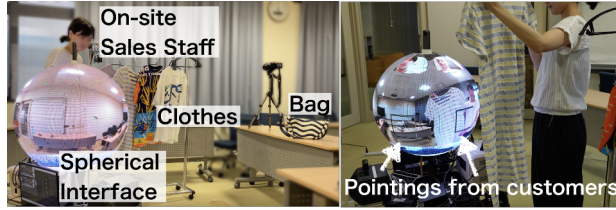


Figure 4: (Left) Demonstration environment, (Right) The faces of remote customers and their points of interest are displayed on the spherical interface, aligned with their respective directions of focus.

clothes (Customer A) and the other as an accompanying person (Customer B). We also prepared the simulated store environment, set up in a room separate from the customers, featured clothes on a hanger rack and a bag placed at some distance from the clothes, as shown in Figure 4 (Left). In the actual operational environment, we intend to utilize a 5G network; however, for this demonstration, we used an indoor Wi-Fi connection.

In this demonstration, we demonstrated one possible scenario with this shopping system. Customer A operated the spherical interface to browse the clothing. Upon finding a particular item of interest, Customer A engaged in conversation with the store staff to get a closer look at the clothing. The staff then showcased the clothing from various angles using the spherical interface to ensure it was clearly visible and made a sales presentation. Meanwhile, remote Customer B, who was browsing in a different direction, pointed at a bag and recommended it to Customer A, as shown in Figure 4 (Right). The store staff, hearing their conversation, displayed the clothing and the bag together, offering a combined view to assist with their decision-making.

4 DISCUSSION AND CONCLUSION

In this paper, we proposed a remote online shopping system that utilizes 360-degree video and a movable spherical display. This system enables multiple online users to engage together in online remote shopping, allowing them to move the camera on the spherical interface and shop online while conversing as if they were physically present at the store. Through the 360-degree video, users can have store staff show them actual products and receive suggestions while shopping. The on-site store staff, in turn, can make product recommendations by observing the facial expressions and focus points of the remote customers.

To validate the concept, we developed a prototype and conducted a demonstration. The demonstration shows that the system has the potential to enable intuitive control for remote users, effective focus alignment with store products, and improved interaction for local sellers through visual feedback and virtual fittings. The system, originally designed for online shopping, has potential applications in street markets, online poster presentations, and virtual tourism, enabling real-time interactions, enhanced engagement, and seamless exploration across diverse contexts. Future work will address the deployment of the system in real-world settings and the collection of user feedback through user studies.

ACKNOWLEDGMENTS

This study was supported by JST ASPIRE JPMJAP2404 and a collaborative research project with Rakuten Mobile, Inc.

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